

Reversible Instance Normalization Experiments

Experiment Guideline

Thesis project

7 August 2026

1 Role of this document

This is the current implementation and experiment contract. The rejected historical ECML/AALTD submission is retained only in the workspace-level parent directory `../../..../latex/submissions/ECML_AALTD_2026_reject`. It may provide historical context, but it is read-only, is not being resubmitted, and does not define the current protocol.

2 Forecasting setting

For user u and query date t , the model observes $X_{u,t} = X_u(t-L, t]$ and forecasts $Y_{u,t} = X_u(t, t+H]$. Chronological regions own target dates. Test1 evaluates future dates for seen users; Test2 evaluates future dates for held-out users. All compared methods share architecture, optimizer budget, batch composition, selected query dates, and original-space MSE evaluation.

3 Normalization components

The location and scale of a lookback may be chosen independently:

$$\ell_x \in \{0, \text{mean}(x), x_L, \text{median}(x)\}, \quad s_x \in \{1, \text{std}(x), \text{MAD}(x)\}.$$

The transformed input is inverted after prediction. The core families are:

Family	Location/scale	Extra parameters
none	none	none
standard	training-global mean/std	none
mean_only	window mean / unit scale	none
scale_only	zero / window std	none
instance	window mean/std	none
median_mad	window median/MAD	none
revin	window mean/std	tied symmetric affine
min	window mean/std	untied global input/output affine

For MIN,

$$\tilde{x} = \gamma \frac{x - \ell_x}{s_x} + \nu, \quad \hat{y} = s_x \left[\alpha \frac{f_\theta(\tilde{x}) - \nu}{\gamma} + \beta \right] + \ell_x,$$

with identity initialization. MIN is global, not user-personalized.

4 Normalized backpropagation

Each applicable forward transform is paired with data-space MSE and normalized MSE:

$$\text{nMSE}(\hat{y}, y \mid x) = \frac{1}{H} \sum_{h=1}^H \left(\frac{\hat{y}_h - y_h}{\text{std}(x) + \varepsilon} \right)^2.$$

Only gradient weighting changes; predictions are denormalized and final tables report original-space MSE. Pairwise comparisons hold seed, model, optimizer, batching, and budget fixed.

5 Centering and transform ablations

Two appendix variants use non-affine instance normalization:

- **revin_last**: center on the last observation instead of the mean;
- **revin_arcsinh**: apply a reversible asinh magnitude compression after instance normalization.

Each is paired with MSE and nMSE training and compared with the matching **instance** control.

6 Seed and uncertainty contract

One seed is the single randomness source for a complete run. It fixes the user partition, data and window sampling, model initialization, and optimization. No separate partition or component seed is introduced. Reported cross-seed dispersion intentionally combines every stochastic source present in the run.

7 Profiles

Mode	Scope	Models	Budget
test	Electricity 504:168, narrow core smoke	PatchTST	seed 1, 2,000 steps
small	Traffic, Electricity, Solar; five settings	PatchTST	seeds 1–3, 10,000 steps
full	nine datasets; five settings	PatchTST	seeds 1–3, 10,000 steps
ultra	full grid	PatchTST and DLinear	seeds 1–3, 10,000 steps

The shared settings are 168:24, 336:48, and 504:168; RevIN also uses 336:96 and 336:720. Publication profiles share outputs only when configuration signatures match.

8 Execution, artifacts, and tables

Each Slurm front owns one family root: `outputs/core`, `outputs/nmse`, `outputs/exotic`, or `outputs/min`. The ordered identity is

`outputs/family/dataset/L_H/backbone/normalization/loss/run_n/seed_n/`

Normalization and loss are separate model configs. Steps, batch size, learning rate, evaluation cadence, split, and stride are pipeline configs; device and Slurm placement are runtime-only. Mode selects delayed path subsets and is not a path or computation-signature field. One seed fixes every stochastic choice.

Each `run_n/manifest.json` uses `schema_version`: 1, lists declared parameters, optional plain provenance, required artifacts, and per-seed state, and has status `not_run`, `running`, `interrupted`, or `completed`. Run identity contains only the schema, identity/model configs, pipeline/experiment configs, and seeds. It never fingerprints or hashes code, Slurm files, data, weights, logs, outputs, or directories; those changes are manual rerun decisions. The schema changes only for a deliberate global contract break, and **new** creates another repeat with unchanged parameters. The default `overwrite_exact` policy skips an identical completion, resumes an identical interruption, and creates the next run for a changed pipeline. Tables support distinct/latest/selected/average pipeline selection and selected/latest/distinct/average repeat selection. Explicit pipeline filters must match even with one run; `SELECTED_RUNS.txt` stores the automatic or pinned selection. Reports live at `outputs/reports/family/mode` and record exact input manifests.

Ninety-four reconstructable historical configurations were migrated. Their source trees remain under `outputs/archive/legacy_`; current readers exclude that archive.

Tables report mean and seed standard deviation, explicit row multipliers, validation-selected policies, and a clearly labeled optimistic test oracle. The oracle is diagnostic only. Dataset configuration uses shared top-level fields, project-scoped overrides, explicit-run precedence, and additive **drop_users**.

9 Practical validation

Run result-format and Slurm workflow tests locally. Training is a remote cluster task; resubmit the same mode to continue from signature-matched seed outputs.